

DAY 14 REPORT

TASK - Seaborn and Graph Creation:

Introduction

Seaborn is a Python data visualization library used to create attractive and statistical graphs easily. It is built on top of Matplotlib and provides simple functions for creating different types of visualizations with less coding.

Import Statement

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Difference Between Matplotlib and Seaborn

Matplotlib	Seaborn
Basic visualization library	Built on Matplotlib
Needs more coding	Needs less coding
Simple graph design	Attractive graph design
Used for full customization	Used for statistical visualization
Manual styling required	Automatic styling available

Graph Creation using Seaborn

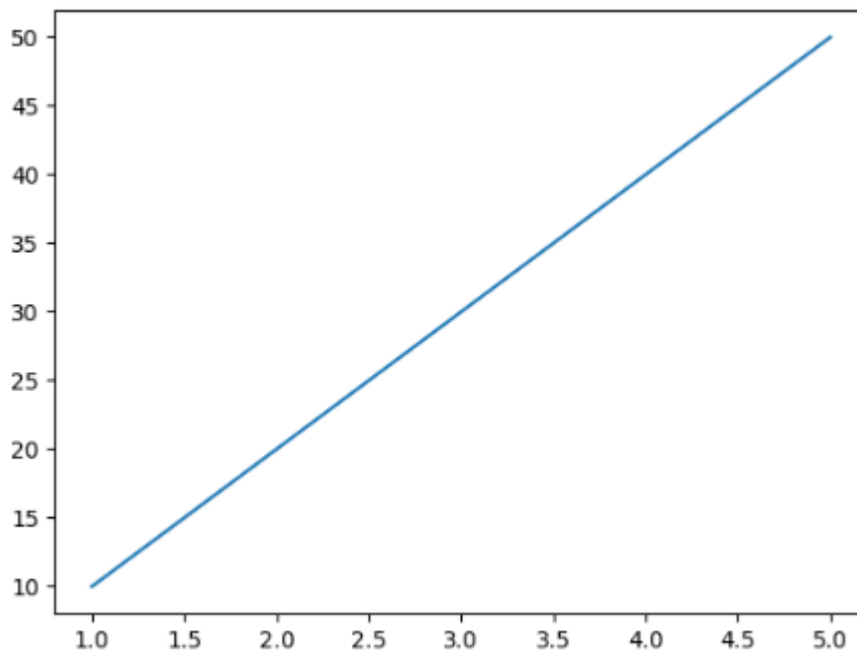
1. Line Plot

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
x=[1,2,3,4,5]
y=[10,20,30,40,50]
```

```
sns.lineplot(x=x,y=y)
plt.show()
```

OUTPUT:



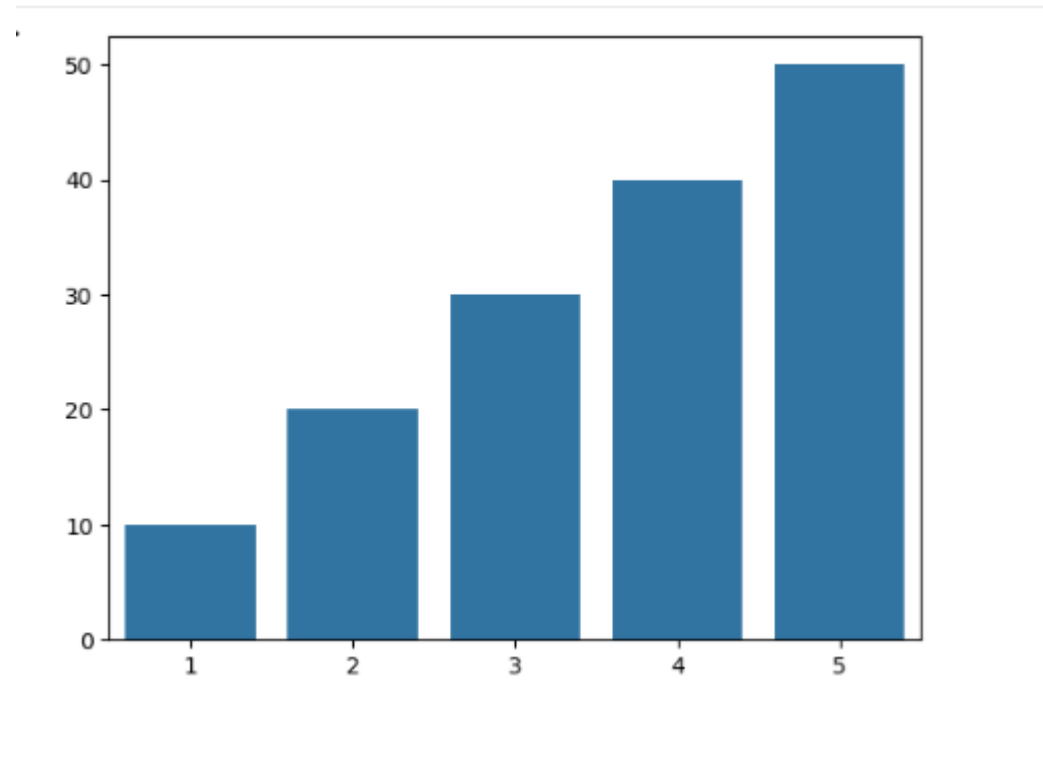
2. Bar Plot

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
x=[1,2,3,4,5]
y=[10,20,30,40,50]
```

```
sns.barplot(x=x,y=y)  
plt.show()
```

OUTPUT:



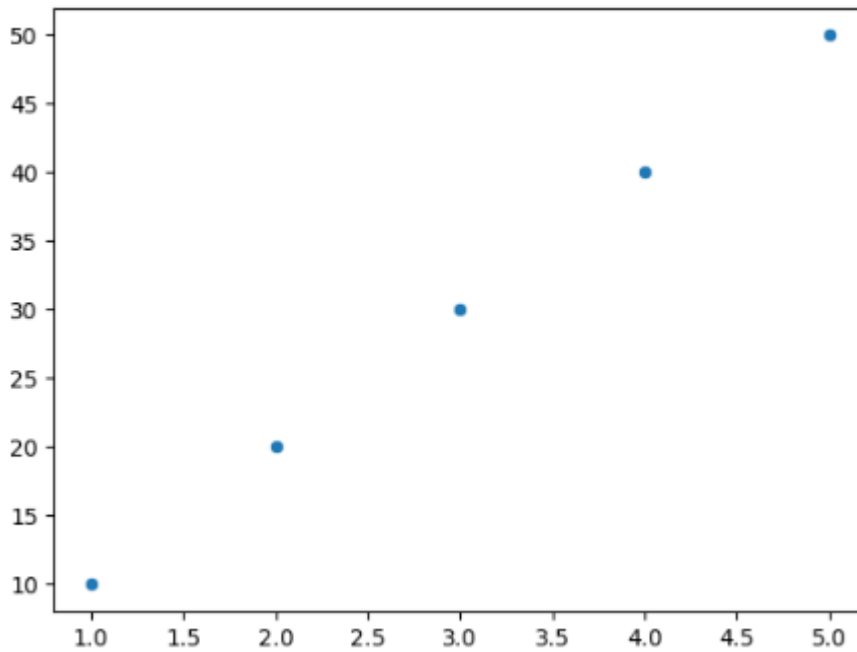
3. Scatter Plot

```
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
x=[1,2,3,4,5]  
y=[10,20,30,40,50]
```

```
sns.scatterplot(x=x,y=y)  
plt.show()
```

OUTPUT:



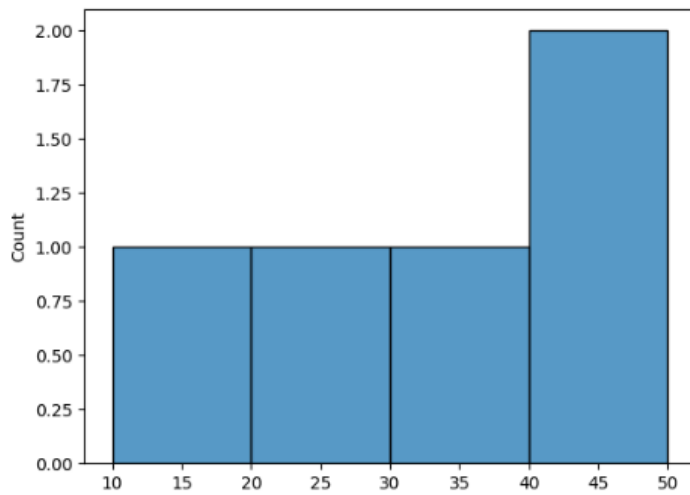
4. Histogram

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
y=[10,20,30,40,50]
```

```
sns.histplot(y)
plt.show()
```

OUTPUT:



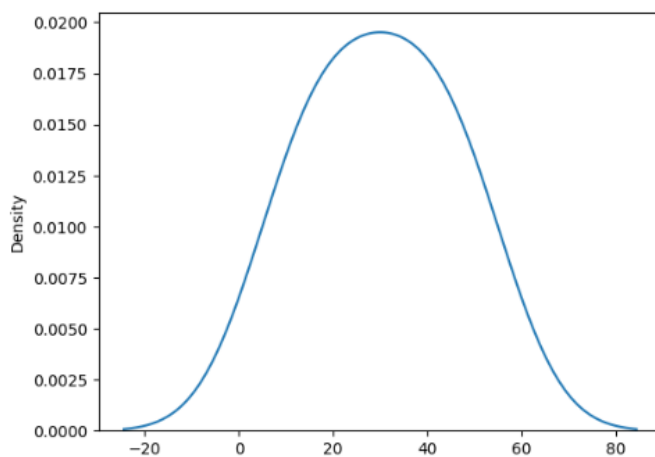
5. KDE Plot

```
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
y=[10,20,30,40,50]
```

```
sns.kdeplot(y)  
plt.show()
```

OUTPUT:



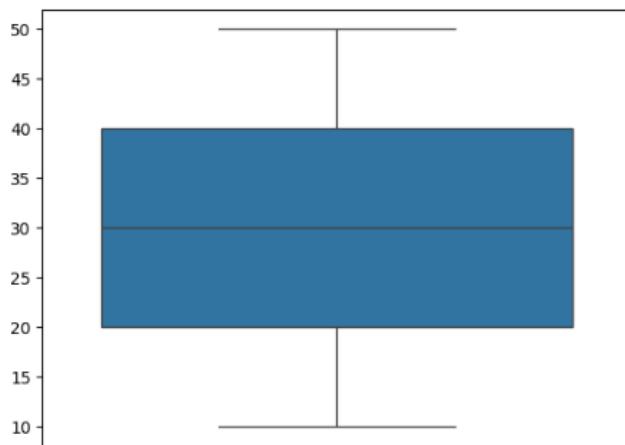
6. Box Plot

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
y=[10,20,30,40,50]
```

```
sns.boxplot(y=y)
plt.show()
```

OUTPUT:

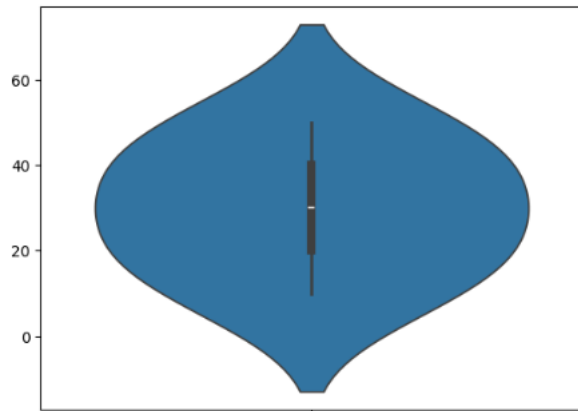


7. Violin Plot

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
y=[10,20,30,40,50]
```

```
sns.violinplot(y=y)
plt.show()
```

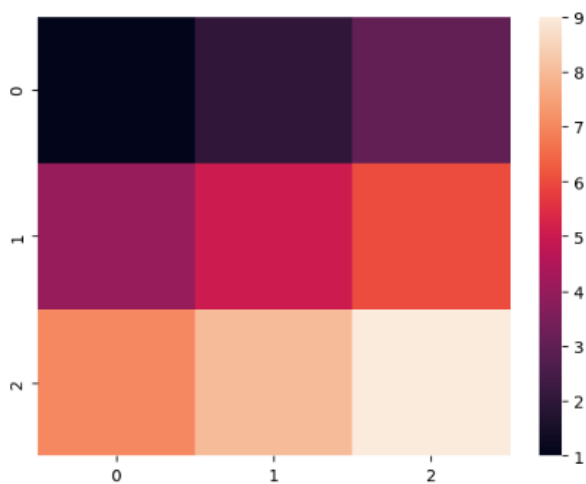


8. Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
data=[[1,2,3],[4,5,6],[7,8,9]]
```

```
sns.heatmap(data)
plt.show()
```



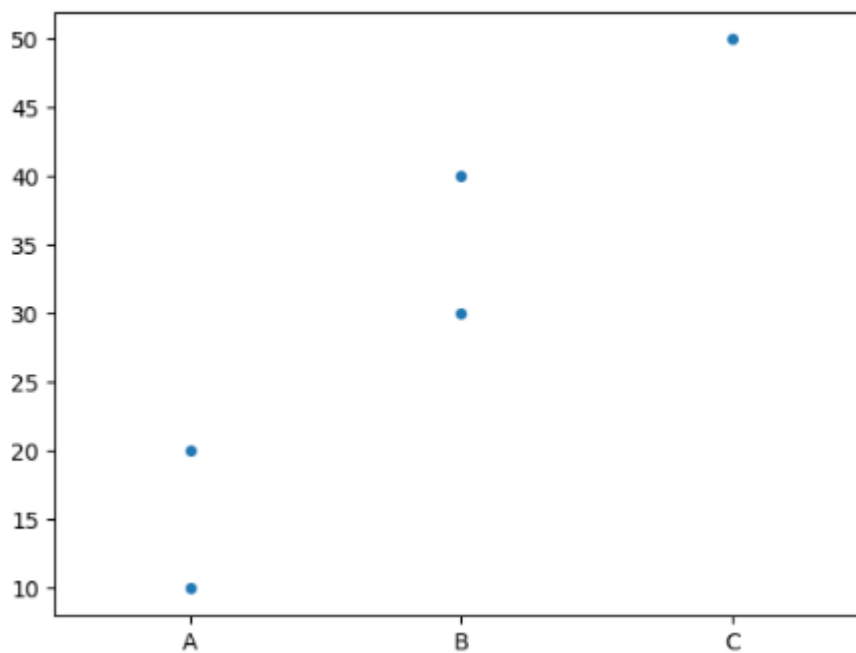
9. Swarm Plot

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
x=['A','A','B','B','C']  
y=[10,20,30,40,50]
```

```
sns.swarmplot(x=x,y=y)  
plt.show()
```

OUTPUT:



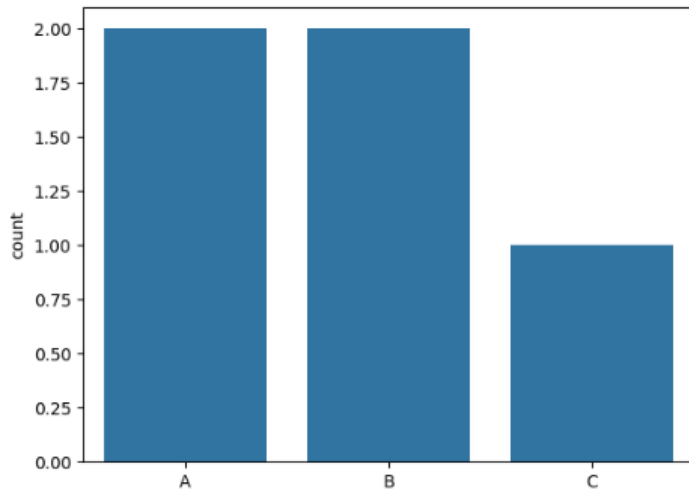
10. Count Plot

```
import seaborn as sns  
import matplotlib.pyplot as plt
```

```
x=['A','B','A','C','B']
```

```
sns.countplot(x=x)  
plt.show()
```

OUTPUT:



11. Joint Plot

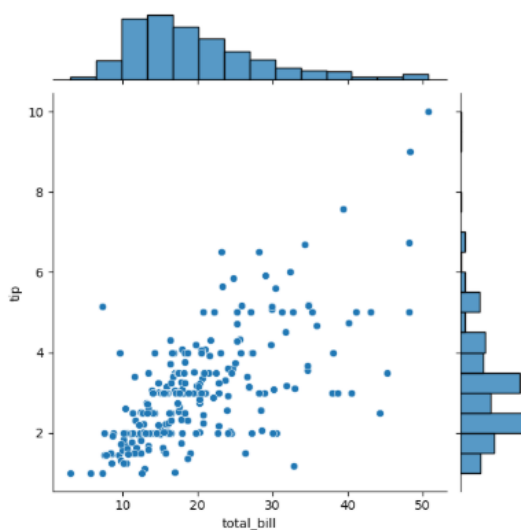
```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips = sns.load_dataset("tips")
```

```
sns.jointplot(x="total_bill", y="tip", data=tips)
```

```
plt.show()
```

OUTPUT:



12.1 Cluster Map

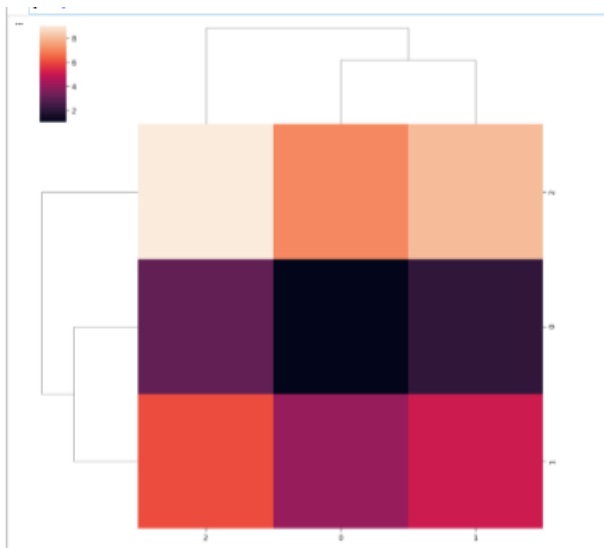
```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
data = [[1,2,3],[4,5,6],[7,8,9]]
```

```
sns.clustermap(data)
```

```
plt.show()
```

OUTPUT:



13.Relational Plot

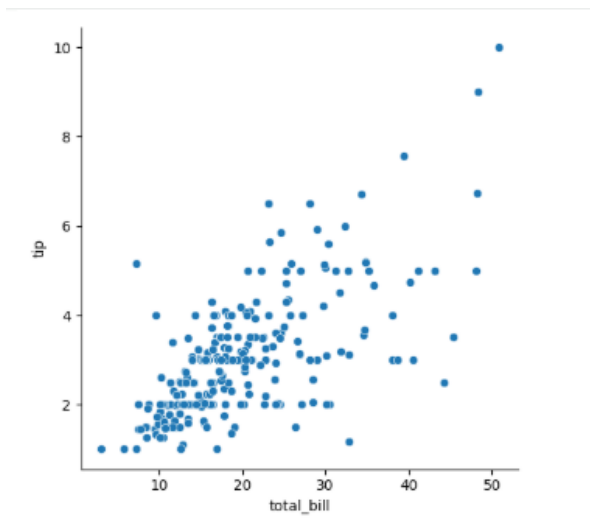
```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips = sns.load_dataset("tips")
```

```
sns.relplot(x="total_bill", y="tip", data=tips)
```

```
plt.show()
```

OUTPUT:



14.Categorical Plot

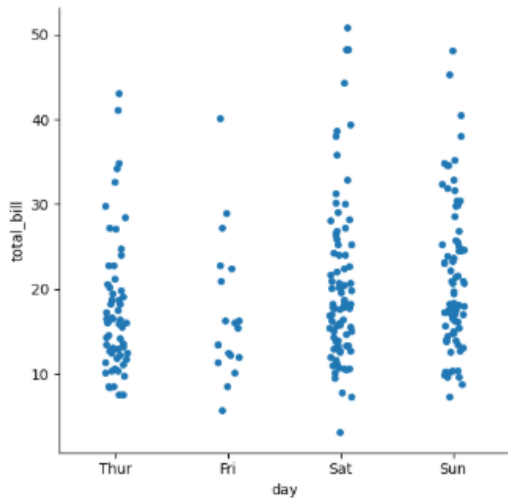
```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips = sns.load_dataset("tips")
```

```
sns.catplot(x="day", y="total_bill", data=tips)
```

```
plt.show()
```

OUTPUT:



15.Distribution Plot

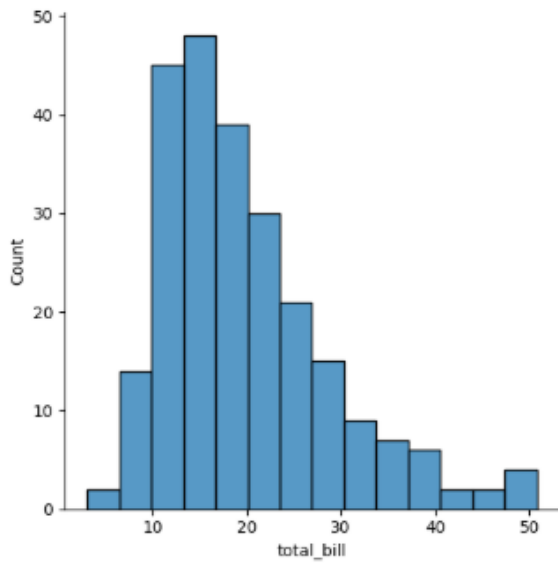
```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips = sns.load_dataset("tips")
```

```
sns.displot(tips["total_bill"])
```

```
plt.show()
```

OUTPUT:



16.PairPlot

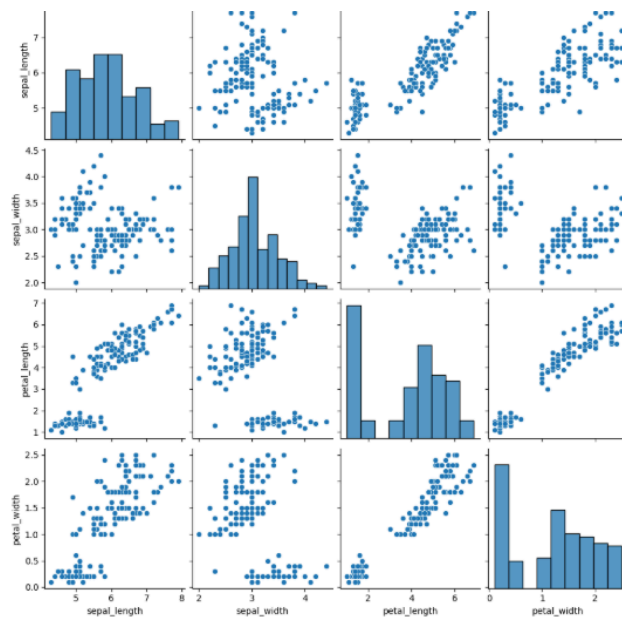
```
import seaborn as sns
```

```
iris=sns.load_dataset("iris")
```

```
sns.pairplot(iris)
```

```
plt.show()
```

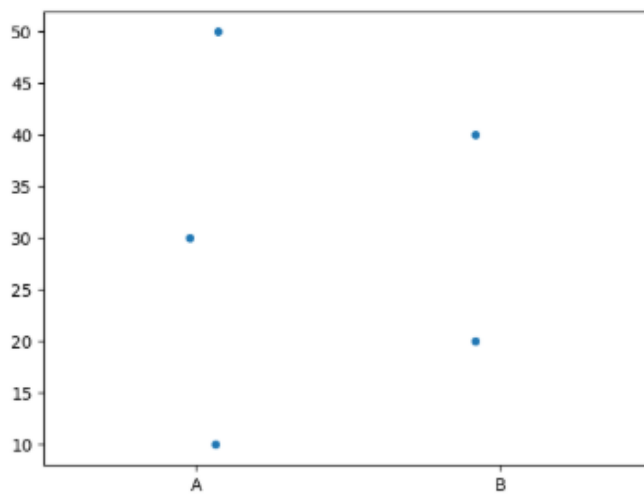
OUTPUT:



17.StripPlot:

```
import seaborn as sns
import matplotlib.pyplot as plt
c=['A','B','A','B','A']
sns.stripplot(x=c,y=y)
plt.show()
```

OUTPUT:



Functions Available in Seaborn

- `sns.lineplot()`
- `sns.barplot()`
- `sns.scatterplot()`
- `sns.histplot()`
- `sns.kdeplot()`
- `sns.boxplot()`
- `sns.violinplot()`
- `sns.countplot()`
- `sns.heatmap()`

- `sns.swarmplot()`
- `sns.stripplot()`
- `sns.pointplot()`
- `sns.regplot()`
- `sns.lmplot()`
- `sns.pairplot()`
- `sns.jointplot()`
- `sns.displot()`
- `sns.relplot()`
- `sns.catplot()`

TASK 2- Bar chart creation using SEABORN

Introduction

Seaborn is a Python visualization library used to create attractive graphs easily. In this task, a bar graph was created using Seaborn to represent student marks visually.

Program

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
x=["A","B","C","D","E","F"]
y=[99,90,45,95,87,65]
```

```
sns.barplot(x=x,y=y,label="Student_Marks")
```

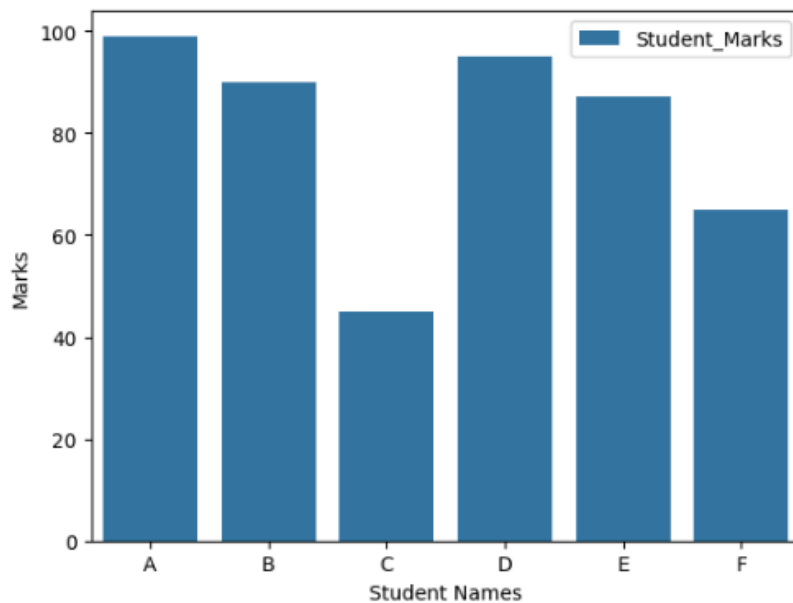
```
plt.legend()
```

```
plt.xlabel("Student Names")  
plt.ylabel("Marks")
```

```
plt.show()
```

Output

A bar graph displaying student names and their marks was generated successfully with labels and legend.



Conclusion

Today, I learned about the Seaborn library and its importance in data visualization. I understood the difference between Matplotlib and Seaborn and created different types of graphs using Seaborn functions in Python.